

GRADUATE MACHINE LEARNING (CS 701)

Lecture Notes: Backpropagation and Computational Graphs

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Instructor: Prof. Turing

1. Executive Overview:

Backpropagation is an efficient application of the multivariable chain rule over directed acyclic computational graphs (DAG).

2. Mathematical Formulation:

Let L be the scalar loss, and $W^{(l)}$ be the weight matrix at layer l .

The forward pass is defined as:

$$Z^{(l)} = W^{(l)} * A^{(l-1)} + b^{(l)}$$

$$A^{(l)} = \text{sigma}(Z^{(l)})$$

By the chain rule, the error term $\delta^{(l)}$ is:

$$\delta^{(l)} = dL / dZ^{(l)} = (W^{(l+1)})^T * \delta^{(l+1)} \odot \text{sigma}'(Z^{(l)})$$

The weight gradient is given by:

$$dL / dW^{(l)} = \delta^{(l)} * (A^{(l-1)})^T$$

3. High-Yield Exam Notes:

- Sliders and matrix dimension checks are guaranteed exam questions.
- Always check that $dL / dW^{(l)}$ matches the exact dimension of $W^{(l)}$.
- Vanishing gradient occurs when $\text{sigma}'(Z)$ approaches 0 for saturated activations like Sigmoid.

4. Practice Check:

Q: Why do we cache intermediate activations $A^{(l-1)}$ during the forward pass?

A: Because they are explicitly required to compute the weight gradient dL/dW during backprop.